Item 01: D_4 symmetrisation stacked with double large primes, k = 4 IC on E(F_{p⁴})

IC improvement round (`../RULES.md`), item 01, 2026-10-14. The protocol, PROTOCOL.md, was frozen before any data was collected. Prior work: `../../ic_next10_20261014/01-fp4-ladder/`.

- **Incumbent:** the D_4 Y-line arm (`d4`).
- **Candidate:** D_4 + double large primes (2LP), tuned on held-apart training curves and then frozen (`d4-2lp`).
- **Also reported:** the untuned stacked arm at Gaudry's |F|^{3/4} rule (`d4-2lp-gaudry`).
- **Rho** appears only as a reference column.

Verdict

REGRESSES at the whole-cost level, at both the constant and the exponent level. The tuned 2LP still loses at the relation phase. Its only gain is in linear algebra.

comparison (test: 24 held-out curves, p = 101…1301)	phase	constant: pooled paired ratio [95 % CI]	verdict	exponent: Δ slope of $\log_2$ ratio per $\log_2$ #E [95 % CI]	verdict
tuned D_4 + 2LP / D_4	**whole**	**1.025 [1.016, 1.037]**	**REGRESSES**	**+0.0055 [+0.0036, +0.0078]**	**REGRESSES**
	relation	1.025 [1.016, 1.035]	REGRESSES	+0.0055 [+0.0033, +0.0079]	REGRESSES
	linear algebra	0.827 [0.802, 0.853]	ADVANCES	−0.0047 [−0.0113, +0.0035]	MATCHES
Gaudry-α D_4 + 2LP / D_4	whole	2.355 [2.043, 2.702]	REGRESSES	+0.0995 [+0.0853, +0.1117]	REGRESSES
	relation	2.355 [2.035, 2.701]	REGRESSES	+0.0995 [+0.0857, +0.1117]	REGRESSES
	linear algebra	0.265 [0.243, 0.293]	ADVANCES	−0.0683 [−0.0821, −0.0549]	ADVANCES

By phase, both stacked arms **TRADE**: linear algebra advances while the relation phase regresses. The relation phase is the whole cost at every measured size: the D_4 LA share is under 10^{-7}. So the whole-cost verdict follows the relation phase.

Fitted handover. This is an extrapolation, not a measurement. Each arm's relation and LA phases are carried on their own fitted power laws.

- Tuned D_4 + 2LP overtakes D_4 only at **#E ≈ 2^128.9**, once LA is about half of D_4's cost.
- The Gaudry-α arm overtakes it at #E ≈ 2^184.3.
- The model also crosses at 2^28.4 and 2^22.5. Both lie at or below the smallest measured size, where the measured tuned ratio is exactly 1.000, so they are fit artefacts.
- The trend line of the paired whole ratio crosses 1 at 2^28.4 and rises from there.
- Modelled tuned ratio: 1.045 at 2^40, 1.146 at 2^64, 1.035 at 2^128 and 0.40 at 2^256.
- The 2^128.9 crossover rests on two things: the measured LA constant (0.83), and an LA exponent difference the data do not resolve (MATCHES). Treat it as indicative only.

Correctness and controls, all passed:

- **Correctness.** Every logarithm of every arm verified, [d]G = Q and equal to the planted d: 120 of 120 on test and 868 of 868 on training.
- **Null A/A** (`d4-2lp-null`, the 2LP path with s = all columns): identical to `d4` in trials, full relations, LA counts and output on 24 of 24 test curves and 14 of 14 training curves.
- **Positive control** (`d4-half`, which discards every odd trial): ratio 2.033 [1.958, 2.117]. That is REGRESSES at the constant level, inside the pre-registered window [1.6, 2.5].
- **Reproduction.** `d4` trial counts equal item 1's committed `ladder.jsonl` on 21 of 21 shared curves.

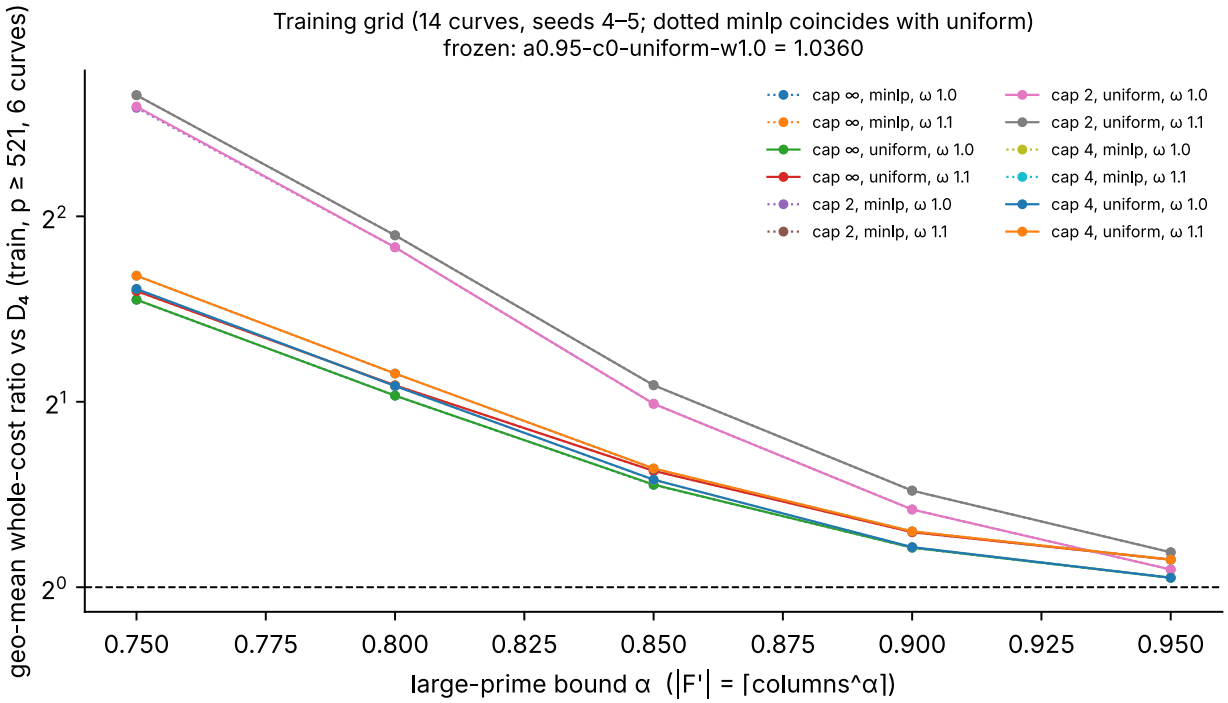
Tuning (training split: seeds 4, 5 × 7 sizes = 14 curves)

The grid has 60 configurations, all run on one shared trial stream per curve:

- large-prime bound α ∈ {0.75, …, 0.95}, where |F'| = [columns^α];
- merge cap (the cycle-finding cut-off) ∈ {∞, 2, 4};
- relation choice ∈ {uniform, minlp};
- relations before the first LA attempt, ω ∈ {1.0, 1.1}.

The objective, fixed in advance, is the geometric-mean whole ratio against D₄ on the six training curves with p ≥ 521. Full ranking: raw/frozen.json.

- **Frozen:** α = 0.95, merge cap ∞ (uncapped), uniform, ω = 1.0. Training score **1.0360**. `minlp` tied it exactly, and the rule kept the first in grid order.
- **Shape of the grid.** The score falls monotonically as α rises. A merge cap of 2 always hurts, a cap of 4 equals uncapped, and ω = 1.1 always hurts.
- **Best below 1:** none. No configuration in the grid scores below 1. The smallest large-prime fraction tried (α = 0.95, about 25 % of columns large) is the best, so the tuning drives 2LP back toward the incumbent.
- The Gaudry rule (α = 0.75) scored 2.93.



training grid

Why a tuned 2LP still loses at the relation phase (measured accounting)

At p = 1033 and 1301 the counts are:

arm	relations used	full relations	pivots	columns in use
D₄	≈ columns + 1	—	—	all
tuned 2LP	≈ columns + 1	≈ s + 1	≈ columns − s	s
Gaudry α	about 0.9 × columns	—	—	—

For example, p = 1033 seed 1: D₄ used 263; tuned used 263 = 200 full + 63 pivots; Gaudry used 241 = 67 + 174.

1. **The large primes are the leftover base columns, not an open-ended prime set.** Nearly every large prime that appears costs one pivot relation. So 2LP cannot cut the number of relations needed below about |F|. It only moves unknowns from the matrix into the pivot store, which is where the 17 % LA saving comes from.
2. **Decompositions with three or four large primes are inadmissible.** A decomposed trial whose decompositions all have ≥ 3 large primes is paid for (C₄) and wasted. That costs the tuned arm 0–10 % more trials, rising with p because the large-prime set grows with p. It costs the Gaudry arm 1.4–3.8×.

Results by size (test split, 3 curves per size)

The tuned and Gaudry columns are paired ratios against D₄ (geometric mean, 95 % bootstrap over the curves at that size).

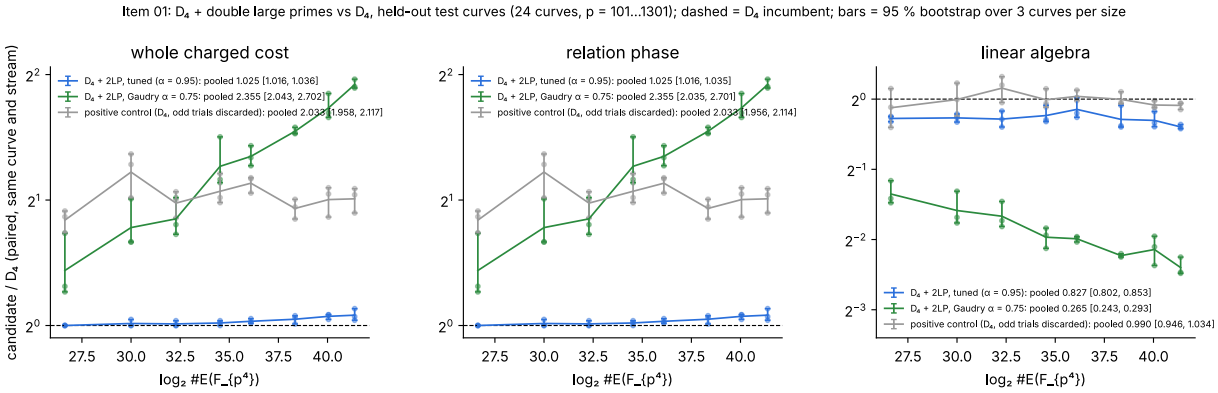
p	log₂ #E	columns	D₄ trials (mean)	tuned whole / D₄	tuned LA / D₄	Gaudry whole / D₄	Gaudry LA / D₄	D₄ S / rho-neg (ref.)
101	26.6	23–24	3,315	1.000 [1.000, 1.000]	0.83	1.36 [1.20, 1.66]	0.39	$1.35 \cdot 10^{8}$
181	30.0	37–50	5,178	1.011 [1.000, 1.034]	0.83	1.72 [1.59, 2.01]	0.33	$7.28 \cdot 10^{7}$

p	log₂ #E	columns	D_4 trials (mean)	tuned whole / D_4	tuned LA / D_4	Gaudry whole / D_4	Gaudry LA / D_4	D_4 S / rho-neg (ref.)
269	32.3	58–69	7,577	1.009 [1.000, 1.027]	0.82	1.80 [1.66, 2.03]	0.32	$5.78 \cdot 10^{7}$
397	34.5	89–107	11,294	1.014 [1.000, 1.027]	0.85	2.41 [2.20, 2.84]	0.26	$3.82 \cdot 10^{7}$
521	36.1	131–134	10,997	1.024 [1.012, 1.039]	0.90	2.55 [2.42, 2.70]	0.25	$2.18 \cdot 10^{7}$
769	38.3	181–202	19,825	1.035 [1.008, 1.055]	0.82	2.92 [2.89, 2.99]	0.21	$2.03 \cdot 10^{7}$
1033	40.1	248–282	24,831	1.052 [1.034, 1.062]	0.81	3.33 [3.16, 3.60]	0.23	$1.36 \cdot 10^{7}$
1301	41.4	311–349	31,024	1.059 [1.030, 1.098]	0.76	3.78 [3.71, 3.90]	0.19	$1.16 \cdot 10^{7}$

Level by level, tuned against D_4:

- **Whole cost.** MATCHES at p = 101–397, where the CI touches 1.000. REGRESSES at every size from 521 up.
- **LA.** ADVANCES at every size.

The rho-negation reference comes from item 1's 16 walks per curve. At p = 1301 it comes from 4 walks per curve run here.



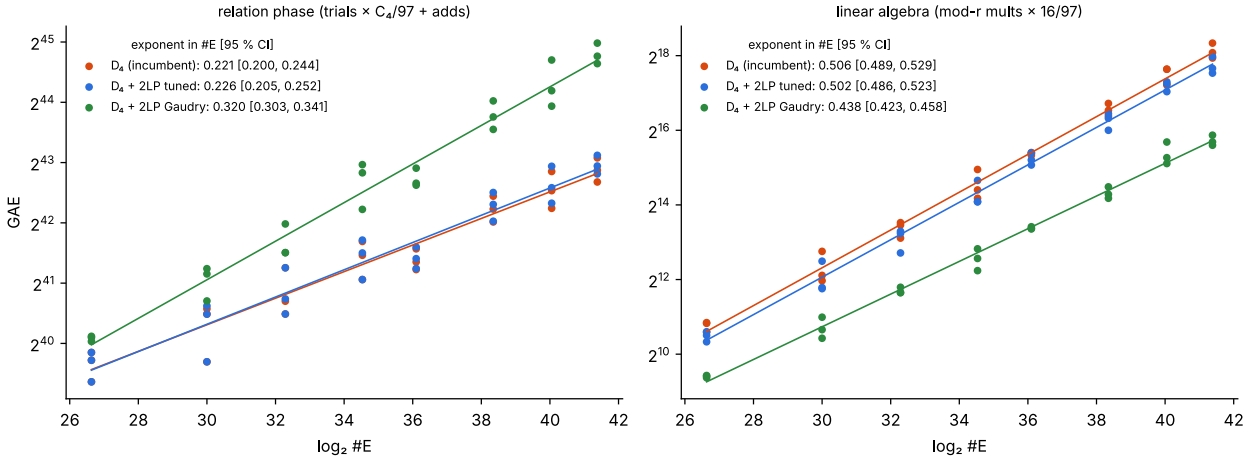
paired ratios

Fitted exponents in #E

OLS of log₂(cost) on log₂ #E over 24 test curves, with 95 % bootstrap over curves.

arm	relation phase	linear algebra	whole
D_4 (incumbent)	0.221 [0.200, 0.244]	0.506 [0.489, 0.529]	0.221 [0.198, 0.244]
D_4 + 2LP tuned	0.226 [0.205, 0.252]	0.502 [0.486, 0.523]	0.226 [0.205, 0.251]
D_4 + 2LP Gaudry α	0.320 [0.303, 0.341]	0.438 [0.423, 0.458]	0.320 [0.303, 0.339]

Item 1's x-line DLP measured a relation exponent of 0.333 and an LA exponent of 0.420. The Gaudry-α stack on the Y-line reproduces that shape: 0.320 and 0.438.



phase costs

ecbench (in-harness reproduction, 4 sizes)

The arm is registered as `ic.k4_gaudry` with `base = yline-d4-lp`, `large_primes = 2` and `lp_exponent = 0.95`. Merge cap ∞, uniform choice and ω = 1 are the harness's existing behaviour, so the frozen configuration is expressible exactly.

- **Session.** `ECBS1hb2883a84d770`, spec ecbench/spec-d4-2lp.json. 4 curves (p = 101, 181, 269, 397, seed 1) × 3 targets × 3 arms = 36 runs, 36 verified.
- **Audit.** `ecbench verify --replay 12`: "audit OK: 36 records, 36 verified, 12 replays reproduced, 0 problems". Receipt `ecbench/session-20261014-d4-2lp.audit.json`, SHA-256 `5380c98c6ae0ff001cce51cdc192afcb86c7fccec57f9c98c31fcd72ce9621fb`.
- **Isolation.** L0, with `--allow-busy --cpus none`. These are operation counts only; no wall-time claim is made.

`ecbench compare` (paired cluster bootstrap, 12 workloads). The harness labels both rows "diagnostic only (unpriced work)": the oracle is uncharged.

B / A	ratio	95 % interval	per curve (p = 101, 181, 269, 397)
ic-k4-d4-2lp / ic-k4-d4	1.0108	[1.001, 1.026]	1.000, 1.020, 1.014, 1.024
ic-k4-d4-2lp-gaudry / ic-k4-d4	1.584	[1.423, 1.807]	1.31, 1.62, 1.84, 2.33

These agree with the ladder at the same sizes: tuned 1.000–1.014, Gaudry 1.36–2.41. Full tables: ecbench/compare.txt.

Deviations from the protocol

1. **Tie in the frozen configuration.** `uniform` and `minlp` tied exactly at the optimum, because with α = 0.95 the two pick rules chose identical relations on every training curve. The protocol named no tie-break; the argmin kept the first in grid order (uniform).
2. **Handover output.** The phase-model handover first reported only the first crossing. After seeing the test output it was changed to list every crossing: the first lay below the measured range and was uninformative. No number in the model changed.
3. **No practical-equivalence margin.** The protocol set none. At p = 101–397 the tuned arm is within 1.5 % of D_4, and the verdicts are stated as measured.
4. **Rho reference at p = 1301.** It uses 4 walks per curve instead of 16. It is a reference only.
5. **Machine.** Not the limit. The p = 1301 curves took about 9 minutes and 1.4 GB each, inside the 3 GB and 60-minute caps. p ≈ 1601 would need about 2.7 GB for the oracle's triple table, too close to the cap, so it was not tried. The whole item took about 80 minutes of single-core time on a shared 4-core host, with load 4–7.

Reproduce

```
        cargo build --release --example ic_improve_01_d4_2lp --bin ecbench
        ./run_train.sh       # 14 training curves, 60 configs + d4 + null
        B=../../../target/release/examples/ic_improve_01_d4_2lp
        $B tune --in raw/train.jsonl --out raw/frozen.json
        ./run_test.sh        # 24 held-out curves: d4, d4-2lp, d4-2lp-gaudry, null, positive control
        $B analyze --in raw/test.jsonl --prior ../../ic_next10_20261014/01-fp4-ladder/raw/ladder.jsonl --out raw/summary.json
        python3 -I plot.py
        # from the repository root
        target/release/ecbench run --wait --allow-busy --cpus none --spec research/ic_improve_20261014/01-d4-2lp/ecbench/spec-d4-
2lp.json --out research/ic_improve_20261014/01-d4-2lp/ecbench/session-20261014-d4-2lp
        target/release/ecbench verify --dir research/ic_improve_20261014/01-d4-2lp/ecbench/session-20261014-d4-2lp --replay 12 --
out research/ic_improve_20261014/01-d4-2lp/ecbench/session-20261014-d4-2lp.audit.json
        target/release/ecbench compare --dir research/ic_improve_20261014/01-d4-2lp/ecbench/session-20261014-d4-2lp --a ic-k4-d4 -
-b ic-k4-d4-2lp
```

Every job ran single-threaded under `nice`, `prlimit --as=3000000000` and `timeout 3600`. Host: Claude Code cloud container, 4 vCPU, 15 GiB, shared by four items. Conductor's control plane was unreachable (`conductor check` refused the connection), so there is no task card.

Files

- New: `examples/ic_improve_01_d4_2lp.rs`, with the `train`, `tune`, `test` and `analyze` subcommands. The oracle, Y-line base and trial stream are copied from item 1.
- Edited, additively: `src/cryptanalysis/ecbench/methods.rs`.
  - It adds the `base` value `yline-d4-lp` to `ic.k4_gaudry`, which allows large primes on the Y-line.
  - The existing refusal now tests the literal `yline-d4`. Behaviour and method hashes of existing specs are unchanged: no parameter was added.
- Data:
  - `raw/train.jsonl` (14 rows) and `raw/frozen.json`;

    - `raw/test.jsonl` (24 rows), `raw/summary.json` and `raw/analyze.stdout`;
    - `ecbench/`.
- Figures: `figures/{paired_ratio,phases,training_grid}.{svg,png}`.
- Report: `REPORT.pdf`, built from this README.